

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Summer Examination – 2025****Course: B. Tech****Branch: Mechanical Engineering / Allied****Semester: III****Subject Code & Name: BTMEC305 & Thermodynamics****Max Marks: 60****Date: 25-07-2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

				(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)				12
1	Which of the following is a point function?				1
	a. Work	CO4	c. Pressure	d. Path	CO2
2	The First Law of Thermodynamics is a statement of:				1
	a. Temperature	CO4	c. Pressure	d. Work	CO3
3	Which of the following processes is isothermal?				1
	a. Constant pressure	CO4	c. Constant volume	d. Constant entropy	CO2
4	Which law states that "entropy of the universe always increases"?				1
	a. Zeroth law	CO4	c. Second law	d. Third law	CO3
5	Which of the following is an extensive property?				1
	a. Pressure	CO4	c. Volume	d. Density	CO1
6	In the absence of heat transfer, the First Law becomes:				1
	a. $Q = W$	CO4	c. $Q = 0$	d. $\Delta U = 0$	CO2
7	Which property remains constant during an isochoric process?				1
	a. Pressure	b. Temperature	c. Volume	d. Density	CO4
8	Which of the following statements is true for an ideal gas?				1
					CO3

	a. Intermolecular forces are very high	b. Volume of gas molecules is significant	c. It follows $PV = nRT$	d. It does not follow the laws of thermodynamics		
9	The Second Law of Thermodynamics introduces the concept of:				CO4	1
	a. Internal energy	CO4	c. Entropy	d. Temperature		
10	In a reversible adiabatic process for an ideal gas, the entropy:				CO5	1
	a. Increases	CO4	c. Remains constant	d. Becomes zero		
11	The Zeroth Law of Thermodynamics is concerned with:				CO3	1
	a. Work transfer	b. Heat transfer	c. Thermal equilibrium	d. Entropy		
12	An ideal gas is defined as one which obeys:				CO4	1
	a. Van der Waals equation	b. Charles' Law	c. Boyle's Law	d. Ideal gas equation		
Q. 2	Solve the following.					12
A)	Explain the Path function and Point function with suitable examples				CO3	6
B)	Describe the following with suitable examples 1) State 2) Change of state 3) Process				CO4	6
Q.3	Solve the following.					12
A)	Explain the First Law of Thermodynamics for a closed system. Derive the energy equation for a non-flow process and discuss its limitations.				CO4	6
B)	Explain the statements of second law of Thermodynamics with suitable examples				CO6	6
Q. 4	Solve Any Two of the following.					12
A)	What is an ideal gas? Derive the ideal gas equation and explain the assumptions made in the derivation.				CO5	6

B)	Discuss the significance of the T-s (temperature-entropy) and P-v (pressure-volume) diagrams in understanding thermodynamic processes.	CO5	6
C)	Explain the concept of internal energy and enthalpy. How are they related to heat and work in thermodynamic systems?	CO4	6
Q.5	Solve Any Two of the following.		12
A)	What is the specific heat of a substance? Explain the difference between specific heat at constant pressure (C_p) and constant volume (C_v), and derive their relationship for an ideal gas.	CO3	6
B)	What is phase change? Describe the phase transformation process of water using the T-v diagram.	CO4	6
C)	What is a property diagram? Explain how Mollier (h-s) and T-s diagrams are used in analyzing thermodynamic processes.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the Separating and Throttling calorimeter with neat sketch	CO4	6
B)	Draw mollier chart and explain its importance	CO6	6
C)	Explain the use of Steam Table for analysis of heat	CO4	6
	*** End ***		